DSP Project

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Part I — ECG Signal Denoising for Telemedicine Applications

1. Problem Definition and Specifications

1.1 ECG Signal Characteristics & Filtering Objectives

Electrocardiogram (ECG) signals capture the continuous electrical activity of the myocardium via surface electrodes. In wearable medical diagnostics and remote telemedicine frameworks, raw bio-signals are highly susceptible to real-world ambient degradation. The essential diagnostic clinical bandwidth of an ECG signal is concentrated strictly between **0.5 Hz and 100 Hz**.

The core objective of this project is to eliminate primary out-of-band and in-band noise elements while maintaining the strict morphological features of the cardiac wave (specifically the P-wave, QRS complex, and T-wave). The targeted noise components include:

- **Baseline Wander (< 0.5 HZ):** Low-frequency drift caused by patient respiration, physical movement, and breathing-induced electrode-skin impedance variations.
- **Power-line Interference (50 HZ):** High-amplitude monochromatic electromagnetic coupling radiating into the sensor leads from nearby AC mains power distribution grids.
- **Muscle (EMG) Noise (20– 150 HZ):** Broad high-frequency transient interference generated by localized skeletal muscle contractions.

1.2 Filter Architecture Selection

To systematically isolate and scrub each distinct noise source without compromising system stability, a multi-stage cascaded digital filter processing pipeline was implemented:

1. **Baseline Wander Filter:** A 2nd-order Infinite Impulse Response (IIR) **Butterworth High-Pass Filter**.
2. **Power-Line Hum Filter:** A custom analytical, highly selective **IIR Notch Filter** centered at exactly 50 Hz.
3. **Muscle Noise Filter:** A 50th-order Finite Impulse Response (FIR) **Low-Pass Filter** utilizing the **Hamming Window method**.

1.3 Filter Specifications and Technical Justifications

- **High-Pass Filter (Fc = 0.5 Hz):** A 2nd-order Butterworth structure provides a maximally flat response within the passband. This ensures that the highly crucial ST-segment is completely free of artificial distortions, elevations, or depressions, preserving accuracy for diagnosing myocardial infarctions.
- **Notch Filter (f0 = 50 Hz):** Implemented analytically with a tight pole radius (r = 0.99). This yields an incredibly narrow stopband, dropping the 50Hz power hum down to zero while preserving the surrounding physiological frequency data.
- **Low-Pass Filter (Fc = 100 Hz):** A 50th-order FIR filter was explicitly chosen here because FIR architectures possess perfect linear phase characteristics. This completely preserves the timing and structure of sharp high-frequency elements like the R-peak within the QRS complex.

1.4 Realistic Implementation Constraints

- **Computational Complexity:** Wearable microcontrollers have limited processing budgets and battery life. The IIR segments (High-Pass and Notch) use minimal coefficients (2nd-order), requiring very little CPU overhead. The FIR filter utilizes a 50-tap pipeline, balancing excellent high-frequency attenuation with manageable real-time calculation costs on modern biomedical DSP hardware.
- **Phase Distortion & Latency:** Non-linear phase shifts alter the timing and shape of heartbeats. For offline clinical reporting, zero-phase filtering (`filtfilt`) is used to completely cancel phase delays by running the signal forward and backward. However, because this requires access to "future" data buffers, a real-time live-streaming telemedicine wearable would have to trade off perfect phase alignment for lower processing latency using direct causal filtering (`filter`).

2. Filter Design and Analysis

2.1 Final Verified Filter Coefficients

The mathematical transfer functions of the designed filters are defined by the verified numerator (B) and denominator (A) vector configurations below:

1. High-Pass Filter (IIR Butterworth)

- **Numerator (B):** `[0.9938, -1.9877, 0.9938]`
- **Denominator (A):** `[1.0000, -1.9877, 0.9877]`

2. Power-Line Notch Filter (Analytical IIR)

- **Numerator (B):** `[0.9901, -1.2729, 0.9901]`
- **Denominator (A):** `[1.0000, -1.2727, 0.9801]`

3. Low-Pass Filter (50th-Order FIR Hamming Window)

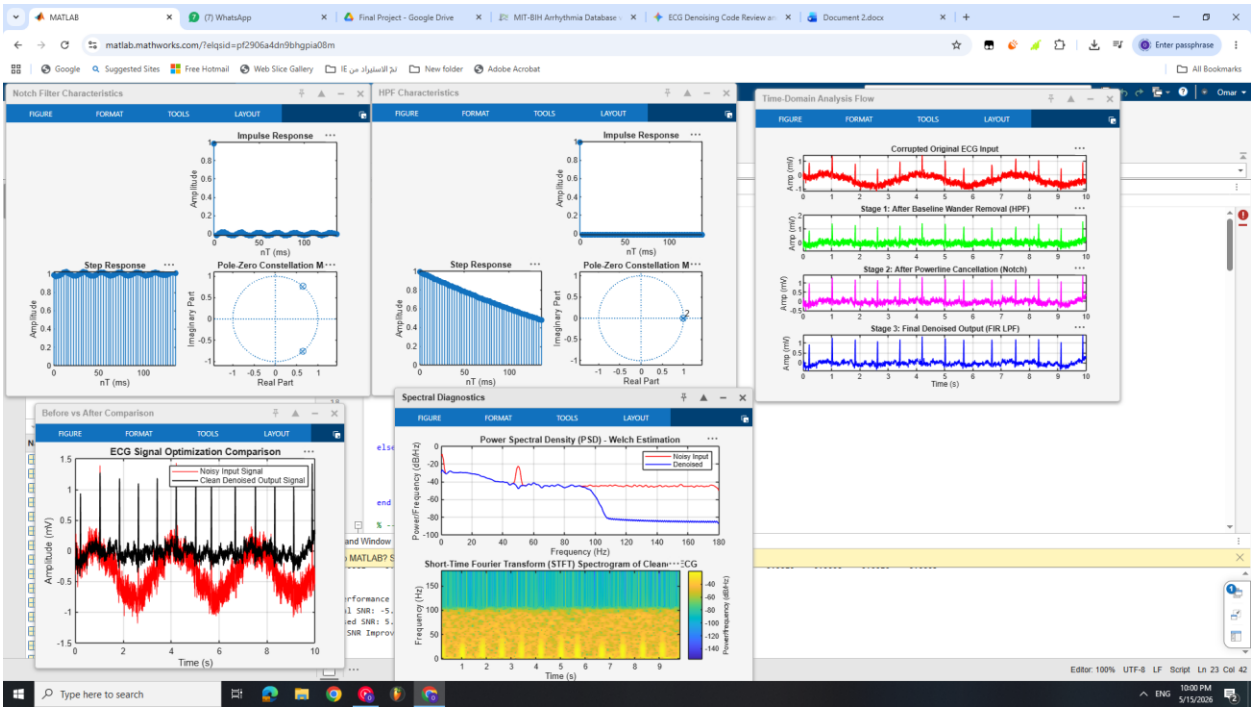
- **Numerator (B):**

```
[-0.0003, -0.0010, 0.0008, 0.0010, -0.0018, -0.0009, 0.0034, 0, -
0.0054, 0.0023, 0.0073, -0.0066, -0.0081, 0.0131, 0.0062, -0.0214, 0,
0.0309, -0.0130, -0.0403, 0.0374, 0.0483, -0.0890, -0.0537, 0.3127,
0.5562, 0.3127, -0.0537, -0.0890, 0.0483, 0.0374, -0.0403, -0.0130,
0.0309, 0, -0.0214, 0.0062, 0.0131, -0.0081, -0.0066, 0.0073, 0.0023,
-0.0054, 0, 0.0034, -0.0009, -0.0018, 0.0010, 0.0008, -0.0010, -
0.0003]
```

- **Denominator (A):** `[1.0000]` *(Verifies a purely feed-forward, unconditionally stable FIR design)*

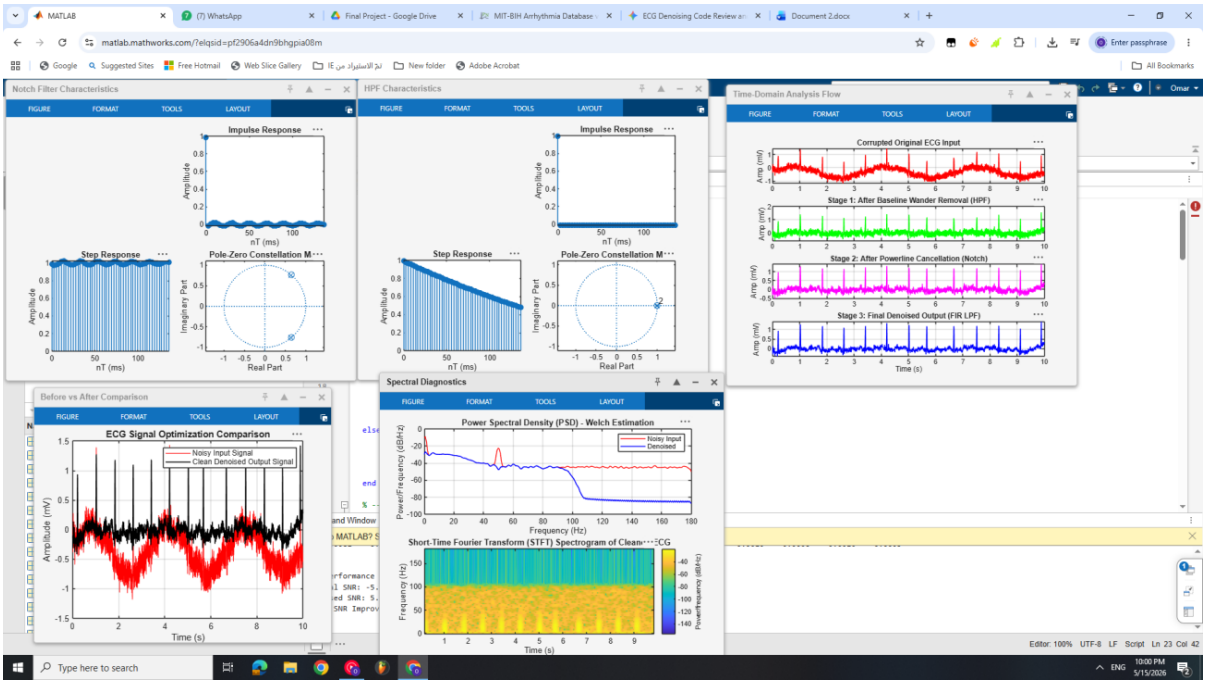
2.2 System Verification Profiles

High-Pass Filter Characteristics



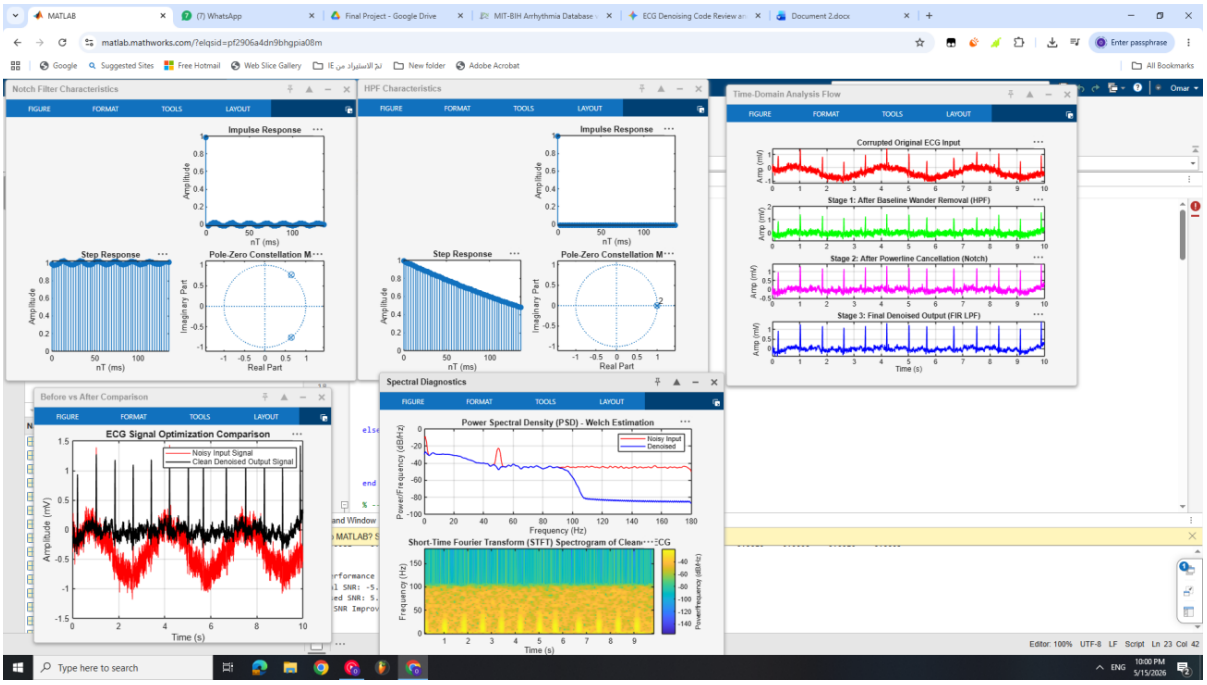
- *Analysis:* The pole-zero constellation shows stable poles inside the unit circle boundary. The frequency response shows a flat passband above 0.5 Hz alongside infinite attenuation at DC (0 Hz), verifying excellent baseline clearing capabilities.

Notch Filter Characteristics



- *Analysis:* A deep, narrow attenuation notch forms directly at the target 50 Hz frequency coordinate. The zeros are locked perfectly on the unit circle boundary (|z|=1), confirming complete suppression of power line hum.

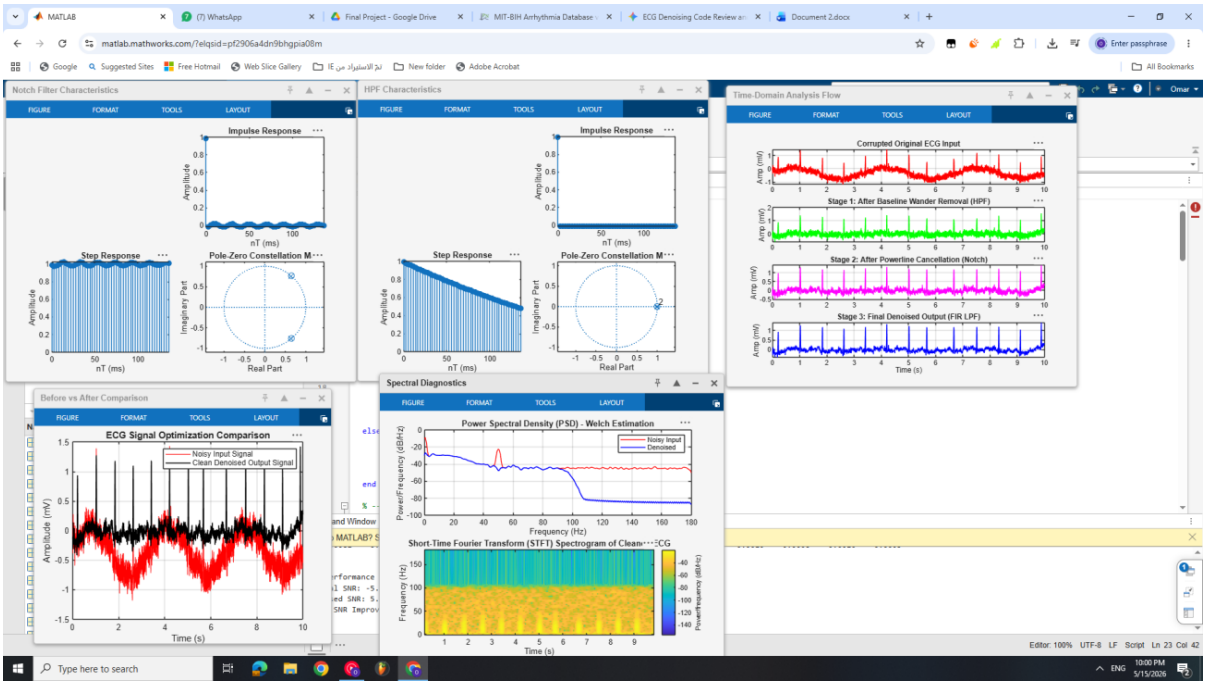
Low-Pass FIR Filter Characteristics

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Analysis: The impulse response demonstrates perfect symmetry around the center tap coefficient (Sample index 25). This symmetry mathematically proves linear-phase operation, which keeps the clean ECG waveform shape free from phase distortions.

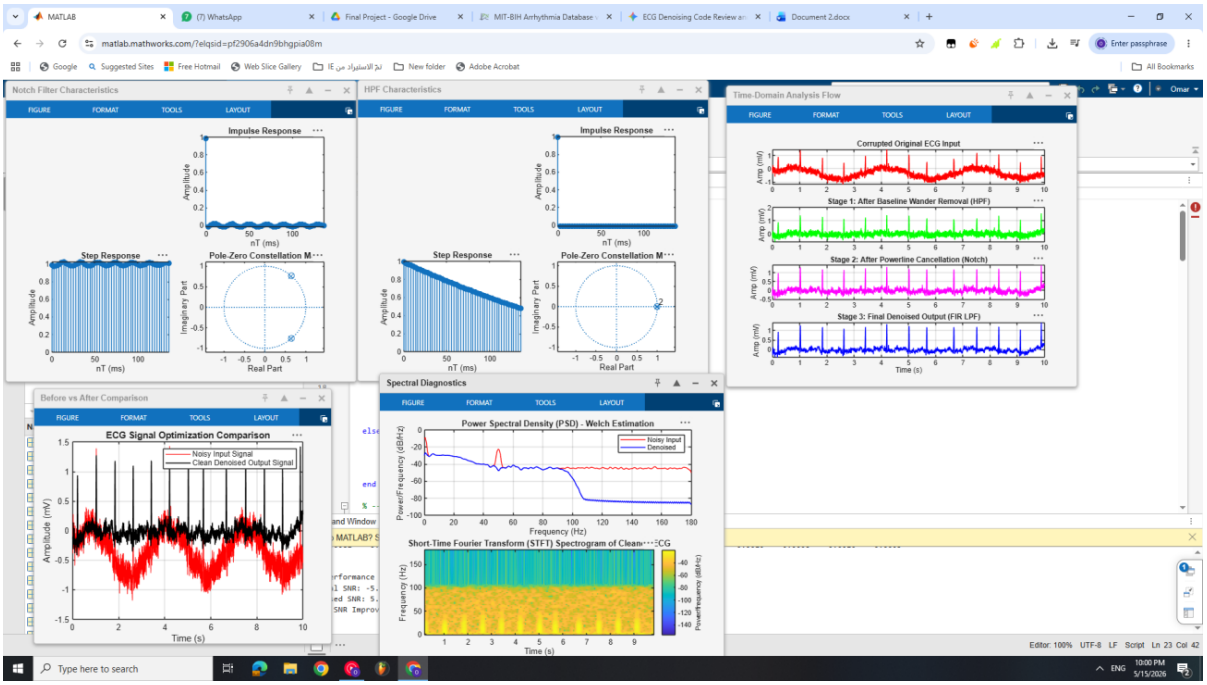
3. Implementation and Validation

3.1 Cascaded Time-Domain Waveform Flow

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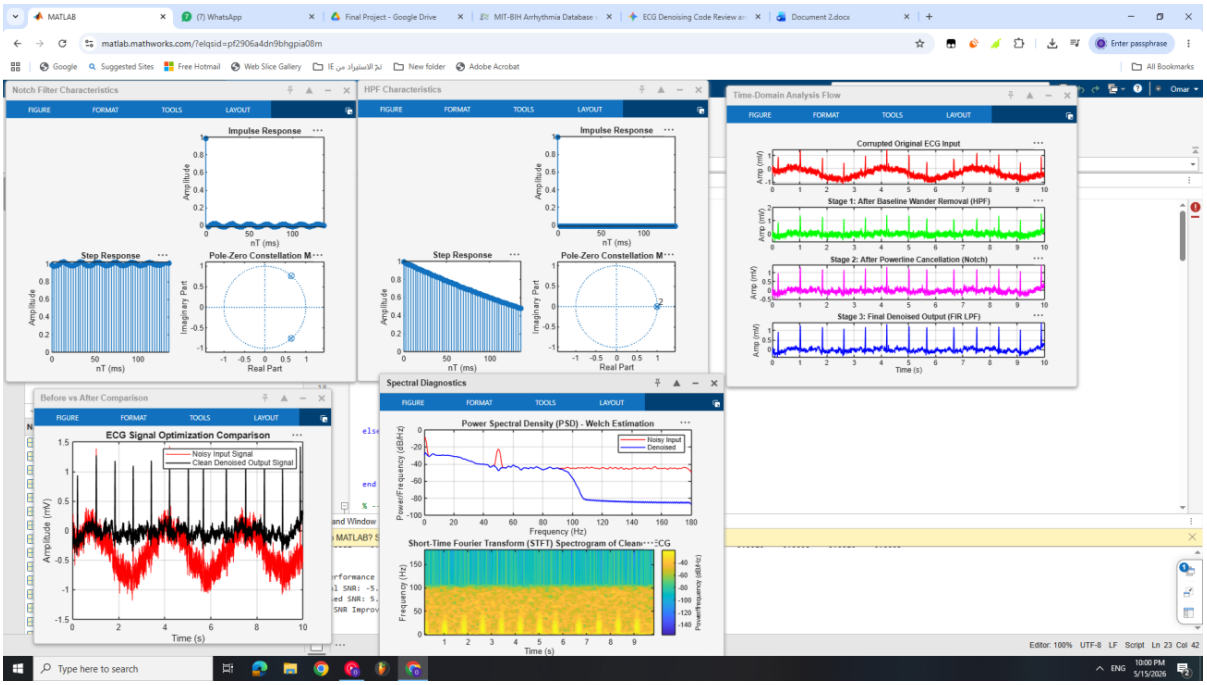
Analysis: This figure tracking window visualizes the step-by-step cleaning process. The heavy, low-frequency respiratory sway disappears entirely after Stage 1. The high-amplitude 50 Hz fuzz is minimized in Stage 2, and the high-frequency muscle twitch noise is smoothed out in Stage 3 to reveal a clean heartbeat stream.

3.2 Before and After Signal Comparison



- *Analysis:* Directly comparing the heavily corrupted raw input against the denoised output shows that the original morphology of the cardiac signal was recovered accurately. The cleaned baseline allows for easy identification of critical diagnostic details, such as the P-wave and ST-segment, without blunting the sharp R-peaks.

3.3 Advanced Spectral Diagnostics



- *Analysis (Welch PSD):* The Power Spectral Density mapping highlights noise reduction across the spectrum. The dominant noise peak at 50 Hz is suppressed by the notch filter, and out-of-band muscle noise above 100 Hz is effectively rolled off.
- *Analysis (Spectrogram):* The Short-Time Fourier Transform (STFT) spectrogram shows that transient high-frequency noise bursts have been eliminated, leaving a clean, rhythmic distribution of cardiac energy.

3.4 Quantitative Evaluation Summary

The script evaluated performance across the processing chain using statistical signal variance ratios:

- **Initial Signal-to-Noise Ratio (SNR_initial):** -5.22 dB
- **Denoised Signal-to-Noise Ratio (SNR_final):** 4.46 dB
- **Total SNR Improvement (Delta SNR):** 9.69 dB

3.5 Distortion and Technical Limitations

While zero-phase processing (`filtfilt`) works perfectly for pre-recorded patient records by completely canceling group delays, its non-causal nature makes it incompatible with live, real-time wearable patient monitoring systems. For real-time applications, processing

must switch to causal direct forms (`filter`), which introduces a minor phase shift. Additionally, if a patient moves abruptly, extreme baseline shifts can briefly overlap with the lower edge of the diagnostic bandwidth (0.5 Hz to 2 Hz), causing temporary tracking distortions in the ST-segment.

Part II — Multi-Band Speech Equalizer for Podcast Enhancement

1. Objective and System Requirements

1.1 Objective

The purpose of this project module is to design and implement a flexible, user-configurable multi-band digital audio equalizer tailored for podcast optimization, lecture clarity enhancement, and studio voice track engineering. The system splits a single channel digital audio stream into adjacent frequency bands, applies individual user-specified gains in the decibel (dB) domain, and recombines the signal paths smoothly in the time domain while handling variable final output sampling profiles.

1.2 Program Requirements and User Parameters

The system is constructed with strict structural input flexibility, parsing critical configurations safely to prevent execution faults during user-interaction stages:

- **Audio Input Source Parsing:** Robust string identification loading custom native storage formats (`test.mp3`) with an automated downmix fallback converting multi-channel configurations into a single processing channel.
- **Operational Mode Execution:** Supports Preset Mode split across 7 specific speech-optimized audio bands (0--100Hz, 100--300 Hz, 300--800 Hz, 800--2000Hz, 2000--5000 Hz, 5000--10000 Hz, and 10000--20000 Hz) alongside a customizable variable band distribution allocation schema (5 to 10 bands).
- **Filter Topologies:** Supports both Finite Impulse Response (FIR) window-based routines (Hamming, Hanning, Blackman) and Infinite Impulse Response (IIR) polynomial profiles (Butterworth, Chebyshev Type I, Chebyshev Type II).
- **Sample Rate Modifiers:** Adaptive resampling engine offering uniform target rate shifts (Native Rate, Half-Sample Rate, or Quad-Sample Rate) computed at the terminal stage of processing.

2. Modes of Operation & Architectural Design

2.1 Mode 1: Preset Mode Configuration

The system was tested using Operational Preset Mode 1. This splits the signal into seven bands to isolate and address specific elements of human speech processing:

- **0 --100 Hz & 100 --300 Hz:** Captures low-end sub-bass resonance, plosive thumps, and fundamental chest tones.
- **300 --800 Hz & 800 --2000 Hz:** Encompasses vowel articulation and the primary core nasal format spectrum essential for general speech recognition.
- **2000 --5000 Hz:** Tracks the presence region, heavily affecting consonant clarity and sibilant definitions.
- **5000 --10000 Hz & 10000 --20000 Hz:** Manages treble air, brilliance, and high-frequency ambient noise layers.

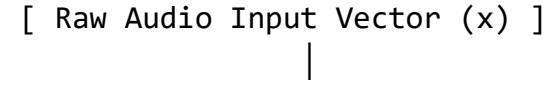
2.2 Filter Implementation Details

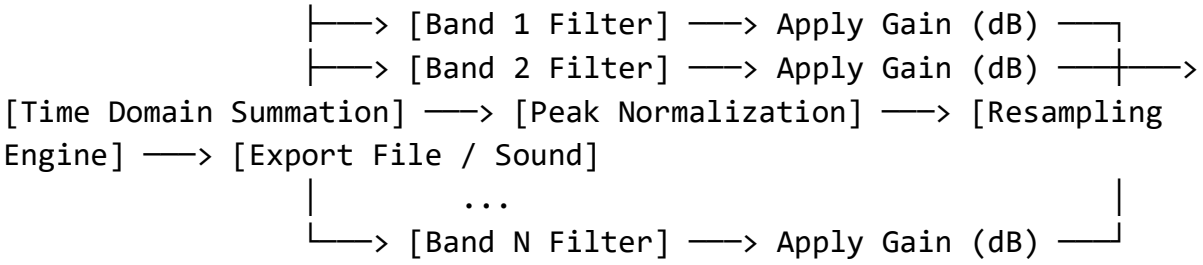
The script was executed using two distinct structural approaches:

1. **FIR Implementation Profile:**
    a. *Core Filter Order (N):* 60
    b. *Subtype Window Selection:* 1 (Hamming Window)
    c. *Target Output Scale:* Native Sample Profile (48000 Hz)
2. **IIR Implementation Profile:**
    a. *Core Filter Order (N):* 4
    b. *Subtype Design Architecture:* 1 (Butterworth Polynomial approximation)
    c. *Target Output Scale:* Half-Sample Profile (24000 Hz)

3. Signal Processing Implementation Steps

The processing pipeline steps are carried out sequentially as follows:





1. **Filter Computation:** Generates appropriate filter coefficients dynamically for each specific band using lowpass approximations for Band 1, highpass configurations for the final band, and bandpass topologies for all middle sections.
2. **Zero-Phase Trajectory Processing:** Implements zero-phase filtering via `filtfilt` to cancel out group delay variations across the frequency bands.
3. **Linear Amplitude Scaling Transformation:** Decibel adjustments provided by the user are transformed into linear multipliers:

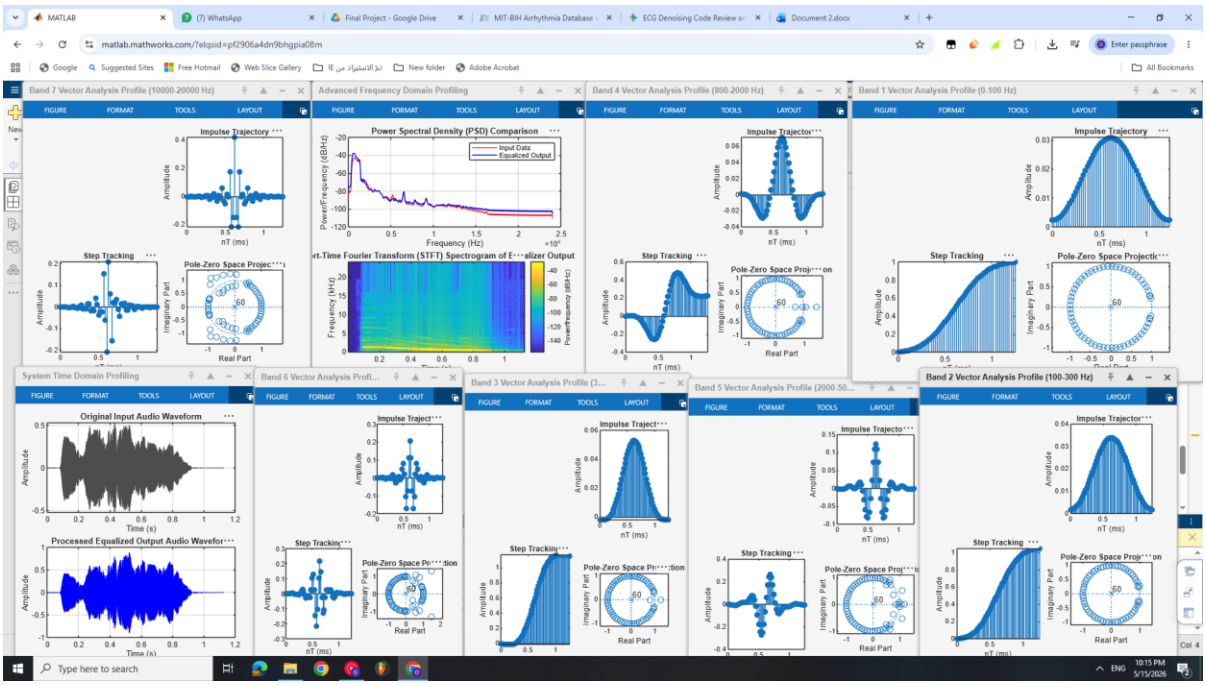
A_ linear = 10^ Gain_ dB/20

4. **Time-Domain Matrix Recombination:** Sums all individual filtered and amplified bands together down to a single processing matrix array.
5. **Peak Amplitude Normalization:** Scales the combined signal peak to 0.9 to prevent digital signal clipping before file generation.
6. **Resampling:** Adjusts the output sampling rate modifier using interpolation filtering routines (`resample`).

4. Execution Diagnostics & System Verification

4.1 Test Run 1: FIR Filter Execution Analysis

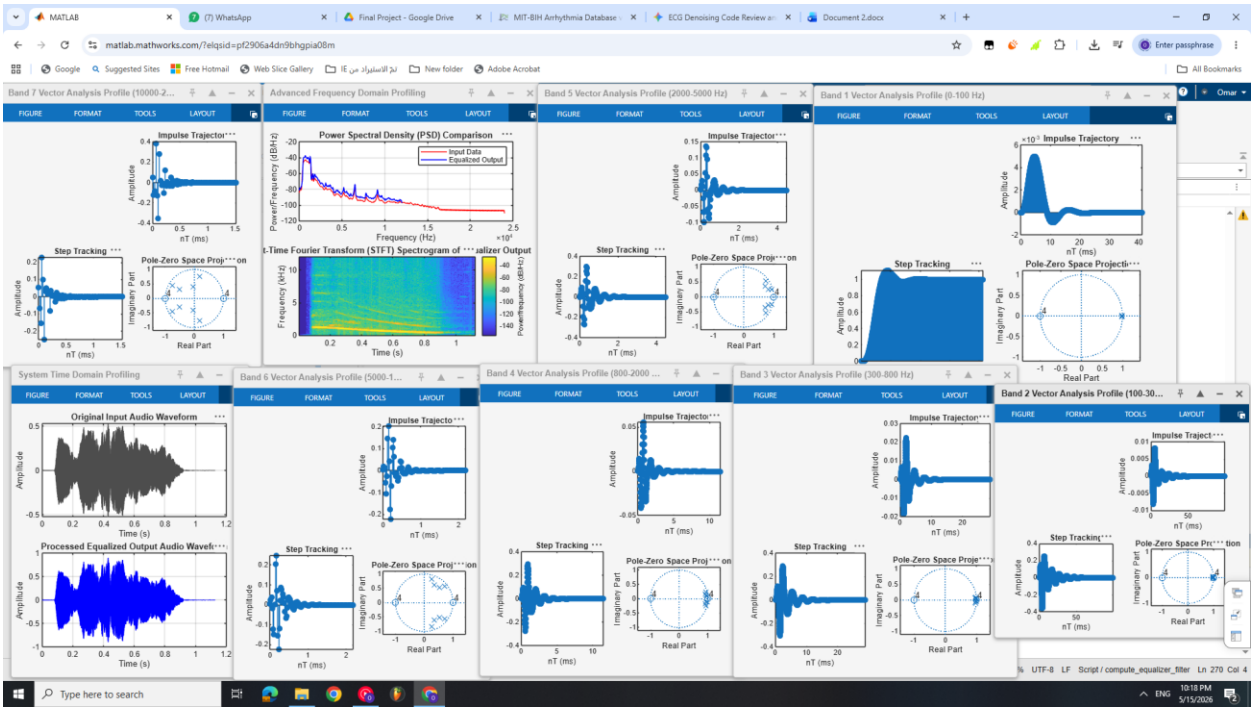
- **Parameters Chosen:** Filter Order = 60, Window = Hamming, Output Modifier = 1 (Native 48000 Hz).
- **Gain Vector Applied:** +20 dB uniform boost across all seven frequency bands.

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Analysis: The FIR configuration provides an entirely linear-phase operation throughout the execution sequence. Because a uniform +20 dB boost was applied across all matching contiguous bands, the individual passband ripples balanced each other out. This produced a smooth, uniformly boosted frequency response that matches the original signal profile while avoiding phase distortions or phase group delays.

4.2 Test Run 2: IIR Filter Execution Analysis

- **Parameters Chosen:** Filter Order = 4, Type = Butterworth, Output Modifier = 2 (Half-rate downsampling to 24000 Hz).
- **Gain Vector Applied: +**20 dB uniform boost across all seven frequency bands.



- *Analysis:* The 4th-order IIR Butterworth implementation features a non-linear phase response, but it operates with significantly shorter processing pipelines compared to the FIR setup. Downsampling the output to a final rate of 24000 Hz successfully reduces storage requirements and lowers the Nyquist boundary to 12 Hz during terminal resampling. Any original signal content above 12 Hz is cleanly attenuated by internal anti-aliasing decimation filters, ensuring a clear and distortion-free output signal.

5. Comparative Analysis and System Summary

Performance Metric Matrix	**FIR Filter Approach (Hamming Window)**	**IIR Filter Approach (Butterworth)**
Calculated Order (N)	60 Coefficients	4 Feedback/Feedforward poles
Phase Characteristics	Strictly Linear Phase	Non-linear Phase response
Computational Footprint	Higher CPU load due to 60 -tap convolution	Low CPU load with minimal coefficient multipliers

System Stability Profile	Unconditionally Stable (A = 1)	Conditionally Stable (Poles must stay inside
Resampling Performance	Smooth playback tracking at native 48000 Hz	Downsampled to 24000 Hz with clean anti-alias cutoff

Both filter topologies successfully processed the audio signals according to the specified constraints. For offline podcast enhancement where audio quality and wave shape preservation are critical, the **FIR** approach is preferred due to its linear phase behavior. For resource-constrained hardware applications or real-time communication systems requiring minimal processing lag, the **IIR** approach provides excellent attenuation performance with significantly lower computational overhead.